

Logistic Regression

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Introduction

- Measures the relationship between the categorical dependent variable and one or more independent variables by estimating probabilities using a logistic function
- Variable we want to predict → dependent variable
- Variables we are using to predict the value of the dependent variable → independent variables

Types of Logistic Regression

- Binomial
 - the dependent variable can have only two possible types
 - dead vs. alive
 - win vs. loss
- Multinomial
 - the dependent variable can have three or more possible types that are not ordered
 - disease A vs. disease B vs. disease C
- Ordinal
 - dependent variables are ordered
 - Win 1, win 2, win 3

How does it work?

- Handles all sorts of relationships, because it applies a non-linear log transformation to the predicted odds ratio
 - Independent variables $\rightarrow$ either continuous or categorical
 - Dependent variables $\rightarrow$ categorical
 - How do we convert a categorical variable into a continuous one?
 - 1) Take the odds of the event happening for different levels of each independent variable
 - 2) Find the ratio of those odds (which is continuous)
 - 3) Take the logarithm of that ratio
- Transformed version of the dependent variable $\rightarrow$ logit
 - logit of success is then fitted to the predictors using linear regression analysis
 - predicted value of the logit is converted back into predicted odds via the inverse of the natural logarithm
- **So although the dependent variable is a categorical variable, the logistic regression estimates the odds (as a continuous variable) that the dependent variable is a success**

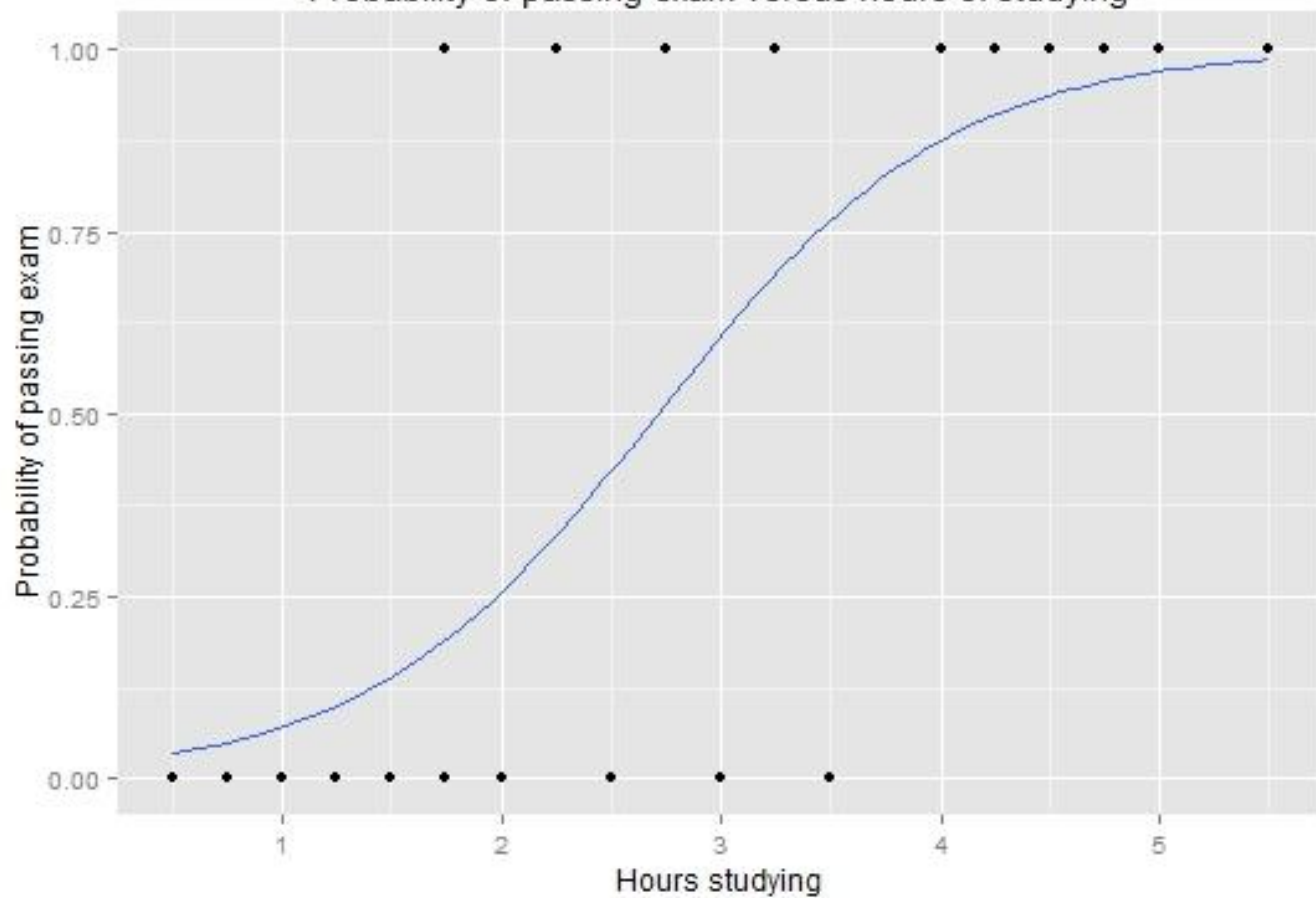
Example

Probability of passing an exam versus hours of study

A group of 20 students spend between 0 and 6 hours studying for an exam. How does the number of hours spent studying affect the probability that the student will pass the exam?

- Why logistic regression? → dependent variable: pass/fail
 - What type? Binomial
- Independent variable → hours spent studying

Probability of passing exam versus hours of studying



Assumptions

- Binary logistic regression requires the dependent variable to be binary and ordinal logistic regression requires the dependent variable to be ordinal
- Since logistic regression assumes that $P(Y=1)$ is probability of the event occurring, it is necessary that the dependent variable is coded accordingly
 - Example for binary regression, the factor level 1 of dependent variable should represent the desired outcome
- Model should be fitted correctly
 - All meaningful variables should be included
 - Good approach to ensure meaningful variables are included is to use a stepwise method to estimate the logistic regression

Assumptions

- Error terms need to be independent
 - Logistic regression requires each observation to be independent
 - Independent variables should be independent from each other
- Linearity relationship of independent variables and log odds
 - Otherwise test underestimates the strength of the relationship and rejects the relationship too easily, that is being not significant (not rejecting the null hypothesis) when it should be significant
- Requires large sample sizes
 - Some suggest at least 30 cases

Examples

Application of Logistic Regression

- Logistical regression is a valuable tool in multiple fields for research and decision making.
 - Medical studies
 - Social Sciences
 - Engineering
 - Marketing
 - Economics



Social Sciences

- Can make predictions on how an individual may vote based on factors including age, income, sex, race, residence, religion, etc.
- Similar factors can be analyzed in predicting crime rate.
- Economically logistic regression can be applied in predicting the likelihood of person's chosen workforce, or to predict the chances of a homeowner defaulting on a mortgage.



Medical Applications



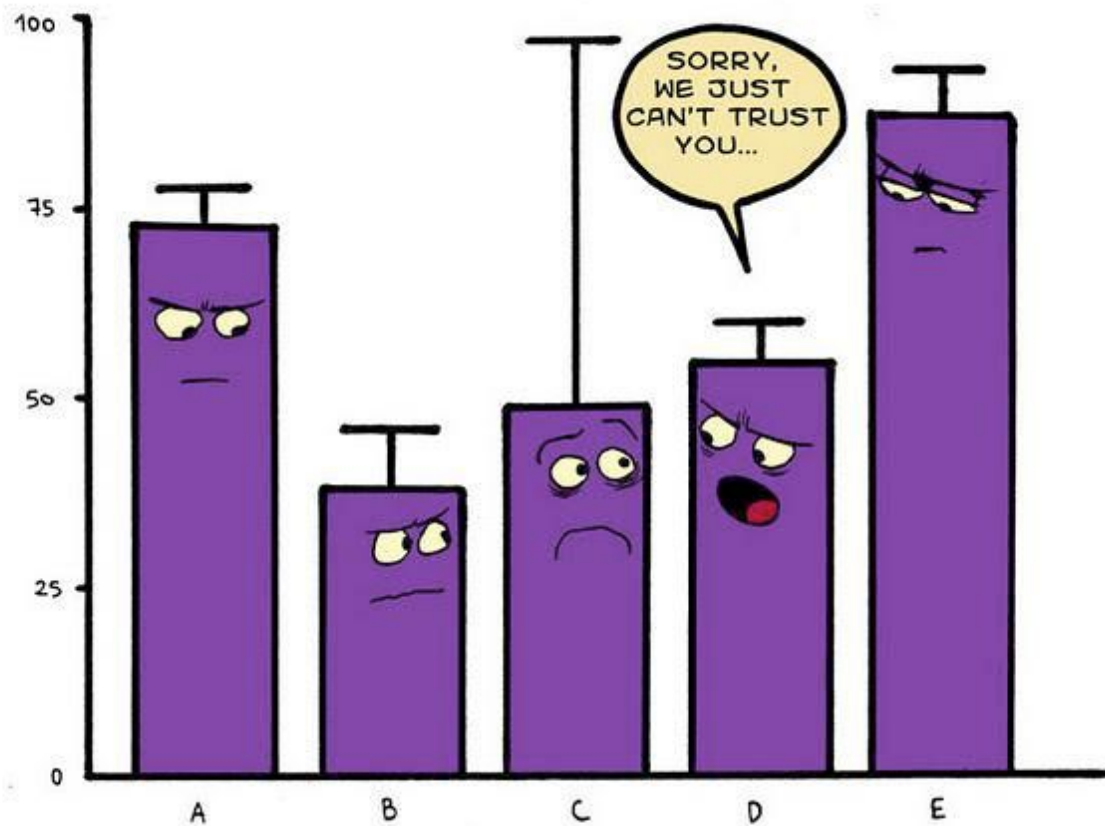
- The TRISS (Trauma and Injury Severity Score) is a model that is widely used to predict mortality in injured patients based on an injury severity score, a revised trauma score and an age index.
- Another common application is in predicting cancer.
 - One study identified 8 independent variable in breast cancer cases.
 - Including patient menopause, first degree relative with breast cancer, family member with other cancer, patient's previous breast trauma history, presence of mass, architectural distortion, skin thickening, and presence of calcification.

Engineering

- An important field that relies heavily on logistical regression are engineers.
- Structural engineers utilize logistic regression in building structures by analyzing materials, strength, support, durability, composition, weather, weight, and height.
- Electrical engineers may also rely on logistic regression when designing an electrical layout dependent on watt supply, watt demand, power needed, wiring composition, and insulation.

Conclusion

- Logistic regression measures the relationship between the categorical dependent variable and one or more independent variables by estimating probabilities using a logistic function
- There are 3 types
 - Binomial
 - Ordinal
 - Multinomial
- Can be applied in many fields
 - Medical
 - Social sciences
 - Engineering
 - Marketing
 - Economics



Sources

- https://en.wikipedia.org/wiki/Logistic_regression
- <https://statistics.laerd.com/spss-tutorials/multiple-regression-using-spss-statistics.php>
- <http://www.statisticssolutions.com/assumptions-of-logistic-regression/>
- http://www.arpapress.com/Volumes/Vol10Issue1/IJRRAS_10_1_02.pdf
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